Quantification: 682_z08
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	11
interior lacuna count	9
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 9)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2599.88	2671.5	1127.64	px^2
major_axis_length_px	lacuna shape	103.523	92.8874	29.9285	px
minor_axis_length_px	lacuna shape	33.6052	36.6597	6.7292	px
aspect_ratio	lacuna shape	3.087	2.9214	0.6438	unitless
eccentricity	lacuna shape	0.9401	0.9395	0.0221	unitless
solidity	lacuna shape	0.8296	0.873	0.1073	unitless
roots_count	headline	12.5556	11	6.0023	count
ring_length_r30_px	headline	454.111	445	220.268	px
ring_length_r60_px	reported	1028.11	965	492.25	px
owned_length_px	ownership-dependent	2820.84	1821.95	2248.43	px
edge_count	ownership-dependent	136.333	97	106.894	count
mean_edge_length_px	ownership-dependent	20.2611	21.0348	2.6495	px
perimeter_px	normalised	253.461	234.894	106.721	px
ring_area_r30_px2	normalised	9783.33	9374	2822.56	px^2
ring_area_r60_px2	normalised	23226.1	23858	6006.58	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9809	1	0.0441	unitless
ring_density_r30	normalised	0.0440315	0.0444469	0.0147704	px^-1
ring_density_r60	normalised	0.0424127	0.042419	0.014898	px^-1
roots_per_100px_perimeter	normalised	5.0992	4.8144	1.4035	per 100 px
ring_attached_length_r30_px	network v2	408.111	390	212.888	px
ring_attached_length_r60_px	network v2	873.556	752	503.418	px
ring_length_w_r30_px	network v2	513.989	487.907	244.88	px
ring_length_w_r60_px	network v2	1162.99	1077.52	548.179	px
sholl_crossings_r10	network v2	10.6667	9	6	count
sholl_crossings_r20	network v2	13.2222	13	5.6298	count
sholl_crossings_r30	network v2	14.8889	13	7.1492	count
ring_width_mean_r30_px	width	6.8931	7.2875	0.8108	px

Per field
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measure	value	units
analysed_area_px2	1.01768e+06	px^2
total_skeleton_length_px	36616	px
canalicular_length_density_per_px	0.0359797	px^-1
junction_count	842	count
junction_density_per_px2	0.00082737	px^-2
lacuna_count	11	count
lacunae_per_px2	1.081e-05	px^-2
skeleton_component_count	273	count

mean_component_length_px	134.125	px
median_component_length_px	18	px
field_length_density_w_per_px	0.0404818	px^-1
field_density_without_flagged_per_px	0.0370742	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	1011	px
unattached_component_count	57	count
canalicular_length_density_attached_per_px	0.0359797	px^-1
field_length_density_with_unattached_per_px	0.0369732	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.5197	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	34950	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	9321	210.577	69.7066	3.0209	0.9436	0.684	1.3112	31.8256	428.162	True
2	2563	90.1026	36.7546	2.4515	0.913	0.9144	1.5381	342.687	87.595	False
3	3496	113.895	40.712	2.7976	0.9339	0.8552	-1.2503	434.254	701.917	False
4	4123	159.751	36.5648	4.369	0.9735	0.7865	1.4743	433.966	210.1	False
5	1521	89.609	30.1271	2.9744	0.9418	0.641	-1.2077	520.501	556.951	False
6	862	61.0771	21.2942	2.8683	0.9373	0.7077	-1.4443	545.484	438.164	False
7	547	50.4784	17.0874	2.9541	0.941	0.8116	-0.4973	672.987	1013.45	True
8	3601	127.976	37.0473	3.4544	0.9572	0.8909	1.4857	713.926	152.006	False
9	1853	92.0913	27.1748	3.3889	0.9555	0.9155	-1.2655	723.731	828.852	False
10	2780	93.6835	39.1668	2.3919	0.9084	0.926	-1.1692	908.613	727.092	False
11								347.049	976.84	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	11	403	1050	4532.36	228	19.8788	5.308
2	12	458	1126	2957.88	153	19.3325	7.5572
3	22	489	1123	3050.05	145	21.0348	7.361
4	22	861	1932	5133.78	275	18.6683	7.38
5	10	409	809	1592.61	73	21.8166	6.856
6	7	445	965	1821.95	97	18.783	6.3276
7	3	166	424	410.458	12	34.2048	5.7406
8	15	657	1479	7513.34	337	22.2948	7.4593
9	11	377	844	1752.39	78	22.4665	7.2875
10	8	334	826	1506.41	65	23.1755	6.8024
11	6	57	149	59.1127	4	14.7782	5.0071

Normalised per lacuna

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lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	769.541	14367	29344	0.6579	0.6229	0.0280504	0.0357824	1.4294
2	234.894	9374	20769	1	0.9593	0.0488585	0.0542154	5.1087
3	313.948	11274	26474	1	1	0.0433741	0.042419	7.0075
4	456.961	14983	32605	1	1	0.0574651	0.0592547	4.8144
5	233.238	9202	19621	1	1	0.0444469	0.0412313	4.2875
6	166.125	7387	17424	1	1	0.060241	0.0553834	4.2137
7	122.604	3917	10590	0.6212	0.5802	0.0423794	0.0400378	2.4469
8	329.522	12020	29668	1	1	0.0546589	0.0498517	4.5521
9	222.752	9104	23858	1	1	0.0414104	0.035376	4.9382
10	245.965	9693	25001	1	1	0.0344579	0.0330387	3.2525
11	77.7401	5013	13615	1	0.8689	0.0113704	0.0109438	7.718

Network v2 per lacuna

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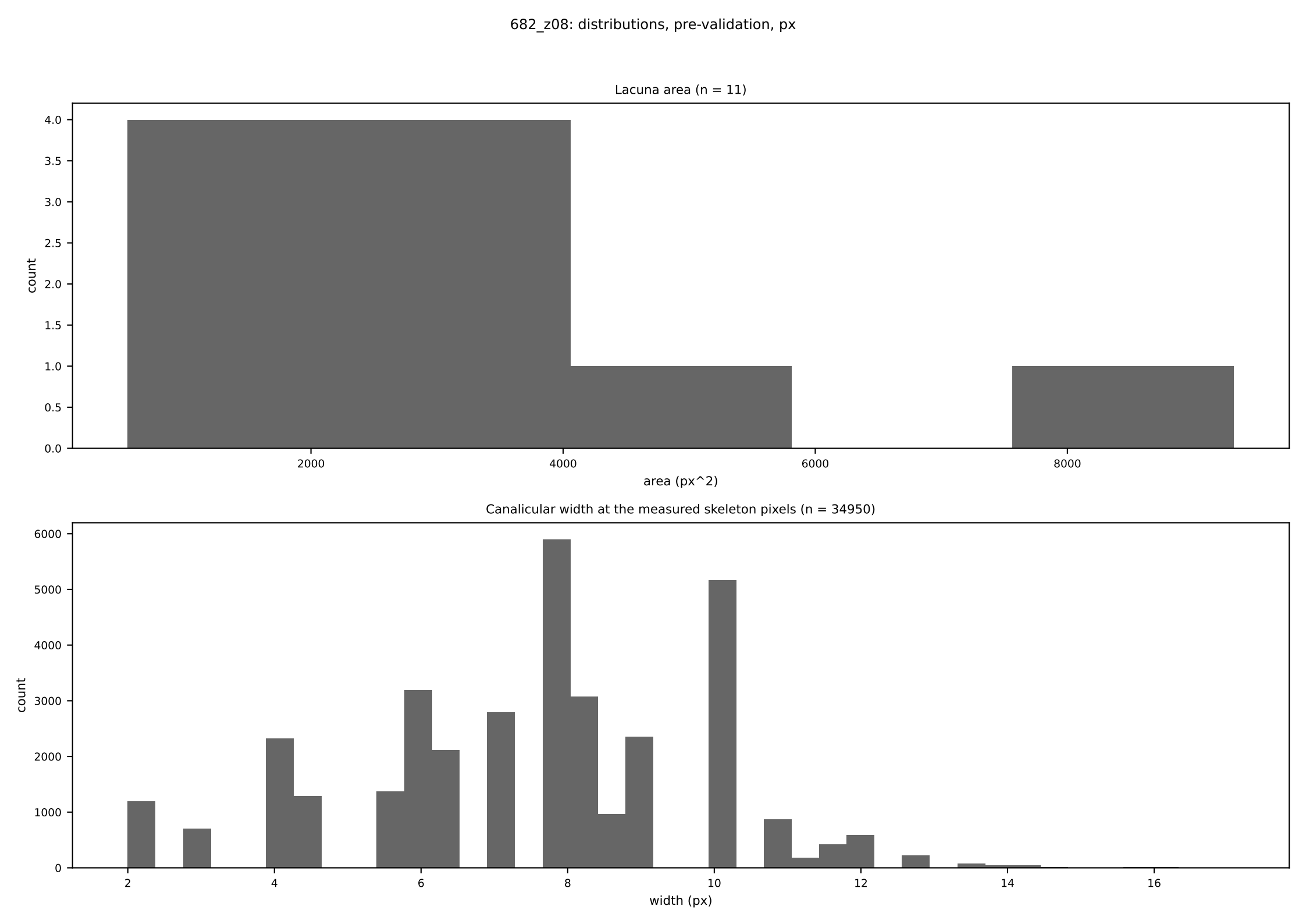
lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	320	687	444.363	1138.68	14	13	12
2	409	1034	515.404	1258.23	12	13	17
3	470	919	557.487	1286.03	14	13	13
4	822	1848	958.583	2170.5	22	22	27
5	358	693	475.617	941.164	6	14	13
6	390	752	487.907	1077.52	9	12	13
7	151	360	194.823	502.085	4	6	7
8	585	1319	744.571	1642.89	15	18	22
9	304	663	435.446	983.703	9	14	15
10	283	582	386.948	942.436	8	12	13
11	52	52	63.9411	164.439	1	1	1


682_z08: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 34950)
count
width (px)